

Project Proposal: Interactive SPH Fluid Simulation with Two-Way Rigid Body Coupling

1. Anawat Bumroongsri 6632245521

2. Sarit Pasiphol 6632234621

1. Motivation and Objective

Fluid simulation is a fundamental challenge in computer graphics, essential for creating realistic environments in games and visual effects. The objective of this project is to develop a real-time, particle-based fluid solver that not only simulates liquid behavior (like splashes and waves) but also enables dynamic interactions with physical objects. By implementing two-way coupling, the fluid will react to moving solids, and the solids will, in turn, be pushed or floated by the fluid.

2. Technical Approach

- **Fluid Simulation (SPH):** We will implement the interactive SPH model proposed by **Müller et al. (2003)**. This method is specifically designed for real-time applications and utilizes specialized kernels (Poly6, Spiky, and Viscosity) to calculate density, pressure, and friction forces efficiently.
- **Two-Way Coupling:** To handle the interaction between fluid and solids, we will use the **Akinci et al. (2012)** method. In this approach, the surfaces of rigid bodies are sampled with "boundary particles." These particles contribute to the fluid's density and pressure, while the fluid exerts hydrodynamic forces back onto the rigid bodies, allowing them to float or be swept away.
- **Acceleration:** To maintain interactive frame rates, a **uniform grid** or **spatial hashing** structure will be implemented for fast neighborhood searches.

3. Project Timeline (4 Weeks)

- **Week 1: Core Framework & Neighborhood Search**
 - Set up the simulation environment and rendering loop (e.g., OpenGL/WebGL).
 - Implement a spatial grid/hashing system to find neighboring particles in $O(1)$ time.
- **Week 2: SPH Solver Implementation**
 - Implement density and pressure calculations using Müller's kernels.
 - Integrate gravity and viscosity forces with a basic time integrator (Euler or Leapfrog).
- **Week 3: Boundary Handling & Two-Way Coupling**
 - Implement particle-based sampling for static boundaries (walls/floor).
 - Introduce dynamic rigid bodies and implement the force exchange (coupling) based on Akinci's method.
- **Week 4: Optimization, Polishing, and Final Report**
 - Tune parameters for stability and "water-like" behavior.

- Add basic visual enhancements (e.g., screen-space fluid rendering or particle coloring).
- Finalize the project report and demonstration.

Key References

SPH Fluid Simulation (Interactive)

- **Müller, M., Charypar, D., & Gross, M. (2003).** *Particle-based fluid simulation for interactive applications*. In Proceedings of the 2003 ACM SIGGRAPH/Eurographics symposium on Computer animation.
 - *Why use it:* This is the classic reference for "interactive" SPH, providing the specific kernels needed for stable, real-time performance.

Two-Way Coupling

- **Akinci, G., Ihmsen, M., Akinci, N., Solenthaler, B., & Teschner, M. (2012).** *Versatile rigid-fluid coupling for incompressible SPH*. ACM Transactions on Graphics (TOG).
 - *Why use it:* This is the modern standard for two-way coupling. It explains how to sample solids with particles to ensure fluid doesn't "leak" through and how to apply forces back to the solids.
- **Becker, M., Ihmsen, M., & Teschner, M. (2009).** *Direct forcing for Lagrangian fluids*. IEEE Transactions on Visualization and Computer Graphics.
 - *Alternative:* This provides a "direct forcing" approach that can be simpler to implement for undergraduate projects requiring robust two-way interaction.

State-of-the-Art Overview

- **Ihmsen, M., Orthmann, J., Solenthaler, B., Kolb, A., & Teschner, M. (2014).** *SPH Fluids in Computer Graphics*. Eurographics State of the Art Report (STAR).
 - *Why use it:* Use this for a comprehensive guide on neighborhood search, time integration, and an overview of different pressure solvers.